10.4 Differentiation Rules

If f is a real valued function of a real variable defined on an open interval I and if $y = f(x)$ is a differentiable function of x, then $\frac{dy}{dx} = f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$. In general finding such direct derivatives using first principle is extremely laborious and difficult operation in the majority of cases. But if we know, once and for all, the derivatives of all the basic elementary functions, together with rules of differentiating the algebra of functions and functions of a function, we can find the derivative of any element – any function without carrying out limit process each time. Hence the operation of differentiation can be made automatic for the class of functions that concern us.

Now we divert our attention to the rules for differentiation of a sum, product and quotient.

Theorem 10.2

The derivative of the sum of two (or more) differentiable functions is equal to the sum of their derivatives. That is, if u and v are two differentiable functions then $\boxed{\frac{d}{dx}(u+v) = \frac{d}{dx}u + \frac{d}{dx}v}$

Proof

Let u and v be two real valued functions defined and differentiable on an open interval $I \subseteq \mathbb{R}$. Let $y = u + v$, then $y = f(x)$ is a function defined on I, and by hypothesis

$$u'(x) = \frac{du}{dx} = \lim_{\Delta x \to 0} \frac{u(x+\Delta x) - u(x)}{\Delta x}$$

$$v'(x) = \frac{dv}{dx} = \lim_{\Delta x \to 0} \frac{v(x+\Delta x) - v(x)}{\Delta x} \text{ exist.}$$

Now, $f(x+\Delta x) = u(x+\Delta x) + v(x+\Delta x)$

$$f(x+\Delta x) - f(x) = u(x+\Delta x) - u(x) + v(x+\Delta x) - v(x).$$

$$\frac{f(x+\Delta x) - f(x)}{\Delta x} = \frac{u(x+\Delta x) - u(x)}{\Delta x} + \frac{v(x+\Delta x) - v(x)}{\Delta x}.$$

This implies, $\lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{u(x+\Delta x) - u(x)}{\Delta x} + \lim_{\Delta x \to 0} \frac{(x+\Delta x) - v(x)}{\Delta x}$.

That is, $\lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = u'(x) + v'(x)$.

That is, $f'(x) = u'(x) + v'(x)$.

or $(u+v)'(x) = u'(x) + v'(x)$.

That is, $\boxed{\frac{d}{dx}(u+v) = \frac{d}{dx}u + \frac{d}{dx}v}$.

This can be extended to finite number of differentiable functions $u_1, u_2, \ldots, u_n$ and $(u_1 + u_2 + \ldots + u_n)' = u_1' + u_2' + \ldots + u_n'$.

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